Lab Handout 1-E
Field-Programmable Gate Array (FPGA) Toolchain Validation: Nexys A7 Loopback Design

Capstone — Module 1: Radio Frequency (RF) Hardware Foundations

1 Objective

Validate the AMD Vivado toolchain and the Digilent Nexys A7-100T board by synthesizing, implementing, and programming a trivial switch-to-LED loopback design. This is the FPGA-side counterpart to Lab Handouts 1-B/1-C and confirms that you can move on to Register-Transfer Level (RTL) development in Module 2.

2 Prerequisites

- No prior lab handouts required for this one; it runs in parallel to the STM32 bring-up track.
- Basic VHSIC Hardware Description Language (VHDL) syntax familiarity (see Module 1 Prerequisites for references).

3 Materials

- Digilent Nexys A7-100T development board (XC7A100T-1CSG324C).
- USB Micro-B cable.
- Host PC with $\geq$50 GB free disk space for Vivado.

4 Procedure

4.1 Step 1 — Install Vivado (first-time only)

1. Download Vivado ML Standard (the free edition that covers Artix-7) from xilinx.com/support/download.html. Create a free AMD account if you don't already have one.
2. Run the installer. When prompted for device families, select **Artix-7** and **7 Series** at minimum; deselecting everything else saves substantial disk space and install time.
3. On Windows, let the installer install the cable drivers when prompted at the end.
4. Launch Vivado once to verify the license. The free Standard edition does not require a license file for Artix-7 devices.

[Figure placeholder]

`LH1E_vivado_installer.png`

*Vivado installer device-family selection screen with **Artix-7** and **7 Series** checked. Deselecting other families saves tens of GB.*

Figure 1: Vivado installer device-family selection screen with **Artix-7** and **7 Series** checked. Deselecting other families saves tens of GB.

4.2 Step 2 — Obtain the master constraint file

1. Download `Nexys-A7-100T-Master.xdc` from Digilent's Nexys A7 reference page (search: "Digilent Nexys A7 master XDC").
2. Save it locally. You will add it to the project and uncomment only the pins you need.

4.3 Step 3 — Create the Vivado project

1. Vivado → *File* → *Project* → *New*. Name the project `nexys_loopback`.
2. Project type: **RTL Project**. Do not specify sources yet.
3. Target device: **xc7a100tcsg324-1**.
4. Finish the wizard.

[Figure placeholder]

`LH1E_part_selection.png`

Vivado "Default Part" dialog with `xc7a100tcsg324-1` highlighted. Any other part will fail implementation on the Nexys A7-100T.

Figure 2: Vivado "Default Part" dialog with `xc7a100tcsg324-1` highlighted. Any other part will fail implementation on the Nexys A7-100T.

4.4 Step 4 — Add the VHDL top module

1. In the *Sources* panel (left side of the Vivado IDE), click the + button (*Add Sources*).
2. Select **Add or create design sources** and click *Next*.
3. Click **Create File**. In the dialog that appears:
   - *File type*: **VHDL**
   - *File name*: `loopback_top`
   - *File location*: leave as <**Local to Project**> (this places the file under `<project>/nexys_loopback.srcs/`

   Click *OK*, then *Finish*.
4. A "Define Module" dialog may appear asking you to define ports graphically — click **OK** to dismiss it without entering anything. You will fill in the port declarations manually in the next step.
5. The new file `loopback_top.vhd` now appears under *Design Sources* in the *Sources* panel. Double-click it to open it in the editor.
6. Replace the entire contents of the file with the following:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity loopback_top is
    Port ( sw  : in  STD_LOGIC_VECTOR(3 downto 0);
           led : out STD_LOGIC_VECTOR(3 downto 0));
end loopback_top;

architecture Behavioral of loopback_top is
begin
    led <= sw;
end Behavioral;
```

Save the file (`Ctrl+S`). Vivado will re-parse the source; the entity `loopback_top` should appear as the top-level design unit in the *Sources* panel with no error markers.

4.5 Step 5 — Add and prune the constraint file

1. In the *Sources* panel, click the + button (*Add Sources*) again.
2. Select **Add or create constraints** and click *Next*.
3. Click **Add Files**, navigate to the `Nexys-A7-100T-Master.xdc` file you saved in Step 2, and select it.

   Important: ensure *Copy sources into project* is checked so the file is copied into the project's `constrs_1` folder. This prevents accidental edits to your master copy.
4. Click *Finish*. The file appears under *Constraints* → *constrs_1* in the *Sources* panel.
5. Double-click `Nexys-A7-100T-Master.xdc` to open it in the editor.
6. The file contains hundreds of commented-out pin assignments. Uncomment *only* the eight lines shown below by removing the leading `#` character on each:

   - `sw[0]`–`sw[3]`: slide switches on pins J15, L16, M13, R15
   - `led[0]`–`led[3]`: LEDs on pins H17, K15, J13, N14

   Double-check the pin identifiers against the master file — different revisions of the XDC occasionally differ. All other lines must remain commented out; leaving extra pins uncommented will cause port-mismatch errors during synthesis.
7. Save the file (`Ctrl+S`).

4.6 Step 6 — Synthesize, implement, generate bitstream

1. In the Flow Navigator, click **Run Synthesis**. Wait for it to complete with zero errors.
2. Click **Run Implementation**. Wait for completion.
3. Click **Generate Bitstream**. This produces `loopback_top.bit`.
4. Open the utilization report. You should see LUT usage in the single-digits out of 63,400 — essentially nothing, which confirms the synthesizer optimized the direct assignment.

[Figure placeholder]

`LH1E_utilization.png`

Post-implementation utilization report. Expect single-digit LUT usage out of 63,400 — the synthesizer optimizes the direct assignment to trivial routing.

Figure 3: Post-implementation utilization report. Expect single-digit LUT usage out of 63,400 — the synthesizer optimizes the direct assignment to trivial routing.

4.7 Step 7 — Program the Nexys A7

1. Connect the Nexys A7 to the host PC via USB. Power the board on. The "FPGA DONE" LED will be off initially.
2. Open the Hardware Manager in Vivado. Click *Open Target* → *Auto Connect*. Vivado should detect the XC7A100T.
3. Right-click the device and select *Program Device*. Point at the `loopback_top.bit` file and click *Program*.
4. After a successful program, the "FPGA DONE" LED should illuminate.

[Figure placeholder]

`LH1E_hw_manager.png`

Vivado Hardware Manager after Auto Connect, showing the detected XC7A100T on the USB-JTAG link.

Figure 4: Vivado Hardware Manager after Auto Connect, showing the detected XC7A100T on the USB-JTAG link.

4.8 Step 8 — Verify physical behavior

1. Flip slide switch `SW0`. LED `LD0` should turn on.

2. Repeat for `SW1`–`SW3` and `LD1`–`LD3`. Each switch must drive exactly one LED.
3. Power-cycle the board. The bitstream is volatile — the LEDs should go dark and the DONE LED should turn off.

[Figure placeholder]

`LH1E_board_leds.jpg`

Nexys A7 with two slide switches up and the corresponding LEDs lit. The DONE LED should also be illuminated.

Figure 5: Nexys A7 with two slide switches up and the corresponding LEDs lit. The DONE LED should also be illuminated.

5 Verification Checklist

- ☐ Vivado project created targeting `xc7a100tcsg324-1`.
- ☐ Synthesis, implementation, and bitstream generation complete with zero errors.
- ☐ Bitstream programs successfully over the Nexys A7 USB-JTAG link.
- ☐ All four slide switches independently control their corresponding LEDs.
- ☐ Power cycle clears the bitstream (expected for volatile configuration).

Common Pitfalls

- **Constraint not found errors**: you edited the wrong `.xdc` file or added it as a generic source rather than a constraints source.
- **Hardware Manager does not see the board**: on Linux, `cable drivers` must be installed manually (see `<Vivado>/data/xicom/cable_drivers/`).
- **Device programs but LEDs don't respond**: the power switch on the Nexys A7 is off, or the board is powered from USB without the jumper set to USB — check the `JP1` / `JP3` jumpers.

Lab Handout 1-E Deliverable

1. VHDL source file `loopback_top.vhd`.
2. Pruned `.xdc` constraints (just the 8 active lines).
3. Screenshot of the Vivado utilization report after successful bitstream generation.
4. Short video or photo sequence showing each switch independently controlling its LED.

Glossary

FPGA	Field-Programmable Gate Array	1
RF	Radio Frequency	1
RTL	Register-Transfer Level	1
VHDL	VHSIC Hardware Description Language	1, 3, 8